

Jacob Rosario

Software Engineer

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Professional Summary

Software Engineer building software to solve real-world problems across web and iOS. Background in biology and competitive volleyball, with hands-on experience in motion sensing, interactive interfaces, and data-driven applications. Brings a scientific approach to problem-solving and an AI-enabled development workflow.

Skills

- **Languages:** JavaScript, Python, Swift, HTML, CSS
- **Frameworks & Libraries:** React, SwiftUI, Three.js
- **Tools & AI:** Claude, ChatGPT, LLMs, Git, GitHub

Projects

RepPower

2026 - Present

- Developing an iOS application that estimates barbell squat velocity from AirPods motion-sensor data for velocity-based training
- Engineered a custom Swift physics pipeline to transform raw motion data into vertical acceleration and velocity estimates while accounting for sensor noise and integration drift
- Implemented zero-velocity drift correction and optimized high-frequency sensor processing to maintain stable real-time measurements without blocking the SwiftUI interface

Volleyball Stat Tracker

Feb 2026

- Developed and deployed a real-time React stat tracker used courtside on iPad across 20+ competitive matches, replacing handwritten note-taking with live performance data for coaches
- Refactored redundant stat sources into a single source of truth, eliminating state drift and supporting tactical decisions that contributed to the team's Division 3 → Division 2 climb

Spike Timer Analyzer

Mar 2026

- Built a React video-analysis tool measuring spike timing from user-marked takeoff, contact, and landing frames, adopted by coached athletes for training feedback
- Chose a lightweight geometric approach over pose-detection ML, running entirely client-side and reaching a 2,000+ follower athletic audience

Personal Website

Jun 2026

- Engineered a responsive 3D portfolio from scratch using React and Three.js, creating custom physics simulations, model rendering, lighting, and interactive environments

Experience

Hustle Volleyball Club

Feb 2026 - May 2026

Sports Analytics Developer & Coach

Burnaby & Vancouver, BC

- Coached a 15-player competitive club through a 4-month season across 5+ tournaments, including provincials and nationals
- Used custom-built stat-tracker data to set lineups, plan practices, and adjust in-game strategy for the club

Vancouver College

Sep 2025 - Dec 2025

Performance Analyst & Assistant Coach

Vancouver, BC

- Analyzed 5+ matches per week to identify opponent tendencies and performance patterns, translating findings into actionable adjustments for players and coaching staff
- Leveraged performance data to inform lineup selections, opponent-specific strategies, and player-development priorities

Starbucks

Sep 2023 - Sep 2025

Barista

Vancouver, BC & San Francisco, CA

- Handled 100+ transactions per shift at high-volume locations, using the POS system to maintain accuracy and efficiency under demand

Education

University of British Columbia

Bachelor of Science (BSc), Biology

Vancouver, BC

- **Achievements:** Dean's List

Certifications

- Anthropic: Claude Code 101
- Anthropic: Claude 101
- Anthropic: AI Fluency: Frameworks & Foundations

Hobbies & Interests

Competitive volleyball, coaching & performance analysis | Sports analytics & biomechanics | AR/VR & 3D technology | Video games | Weightlifting & Olympic lifting | Content creation & video editing | Specialty coffee | Anime & manga | Playing guitar